Exp 17: EXPERIMENT TO UNDERSTAND LINEAR REGRESSION FOR A GIVEN DATA SET

AIM: Understand the Linear regression for the dataset given.

SAMPLE CODE:

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import pickle

# Load the dataset
df = pd.read_csv('Salary_data.csv')

# Display dataset information
print(df)

print(df.info())

# Remove missing values
df.dropna(inplace=True)

# Display dataset information again
print(df.info())
```

```
# Statistical summary
print(df.describe())

# Separate features and labels
features = df.iloc[:, [0]].values
label = df.iloc[:, 1].values

# Split dataset into training and testing sets
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=0
)

# Create and train the Linear Regression model
model = LinearRegression()
model.fit(x_train, y_train)

# Evaluate the model
print("Training Score:", model.score(x_train, y_train))
print("Testing Score:", model.score(x_test, y_test))

# Display model coefficients and intercept
print("Coefficient:", model.coef_)
print("Intercept:", model.intercept_)

# Save the trained model
with open('SalaryPred.model', 'wb') as file:
    pickle.dump(model, file)
```

```
# Load the saved model
with open('SalaryPred.model', 'rb') as file:
    model = pickle.load(file)

# Predict salary for user input
yr_of_exp = float(input("Enter Years of Experience: "))
yr_of_exp_np = np.array([[yr_of_exp]])

salary = model.predict(yr_of_exp_np)

# Display predicted salary
print(
    "Estimated Salary for {} years of experience is: {}".format(
        yr_of_exp, salary[0]
    )
)
```

OUTPUT:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

```
df.dropna(inplace=True)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

```
df.describe()
```

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

```
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
```

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_st
```

```
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
```

```
▾ LinearRegression

LinearRegression()
```

```
model.score(x_train,y_train)
```

```
0.9603182547438908
```

```
model.score(x_test,y_test)
```

```
0.9184170849214232
```

```
model.coef_
```

```
array([[9281.30847068]])
```

```
model.intercept_
```

```
array([27166.73682891])
```

```
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
```

```
model=pickle.load(open('SalaryPred.model','rb'))
```

```
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
```

```
Enter Years of Experience: 44
```

```
print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary)
```

```
Estimated Salary for 44.0 years of experience is [[435544.30953887]]:
```

RESULT: Thus, experiment to understand linear regression for a given data set is conducted.